

CFD Notes

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1 General Conservation Law

The finite volume method is fundamentally designed for partial differential equations written in conservative form. Consider the general conservation law

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{F}(u) = S(u) \quad (1)$$

where u is the conserved variable, $\mathbf{F}(u)$ is the flux vector, and $S(u)$ is a source term. The key idea of the finite volume method is that conservation is enforced over a control volume rather than at a single point.

2 Control Volume Formulation

Let V_i denote a control volume with boundary ∂V_i . Integrating the conservation law over the control volume gives

$$\int_{V_i} \frac{\partial u}{\partial t} dV + \int_{V_i} \nabla \cdot \mathbf{F} dV = \int_{V_i} S dV \quad (2)$$

The first term represents accumulation inside the control volume, while the second term represents net flux through the boundary.

3 Application of the Divergence Theorem

The divergence theorem converts the volume integral of the divergence term into a surface integral:

$$\int_V \nabla \cdot \mathbf{F} dV = \oint_{\partial V} \mathbf{F} \cdot \mathbf{n} dS \quad (3)$$

where $\mathbf{n}$ is the outward unit normal vector. Applying the divergence theorem to the conservation law gives

$$\frac{d}{dt} \int_{V_i} u dV + \oint_{\partial V_i} \mathbf{F} \cdot \mathbf{n} dS = \int_{V_i} S dV \quad (4)$$

This equation is the continuous finite volume formulation.

4 Cell Average

Define the cell average by

$$\bar{u}_i = \frac{1}{|V_i|} \int_{V_i} u dV \quad (5)$$

where $|V_i|$ is the volume of the control volume. The conservation equation becomes

$$\frac{d}{dt} (\bar{u}_i |V_i|) + \oint_{\partial V_i} \mathbf{F} \cdot \mathbf{n} dS = \int_{V_i} S dV \quad (6)$$

This equation expresses the fact that the rate of change of conserved quantity inside a control volume is determined by the net flux across the boundary and any source contribution.

5 Discrete Surface Fluxes

The surface integral is approximated as a sum over the faces of the control volume. Suppose the control volume has N_f faces. Then

$$\oint_{\partial V_i} \mathbf{F} \cdot \mathbf{n} dS \approx \sum_{f=1}^{N_f} (\mathbf{F} \cdot \mathbf{n})_f A_f \quad (7)$$

where A_f is the area of face f . The semi-discrete finite volume formulation is therefore

$$\frac{d\bar{u}_i}{dt} = -\frac{1}{|V_i|} \sum_{f=1}^{N_f} (\mathbf{F} \cdot \mathbf{n})_f A_f + \bar{S}_i \quad (8)$$

This is the general finite volume equation.

6 Specialization to One Dimension

In one dimension, the conservation law becomes

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = S(u) \quad (9)$$

Consider the control volume

$$[x_{i-1/2}, x_{i+1/2}] \quad (10)$$

with width

$$\Delta x = x_{i+1/2} - x_{i-1/2} \quad (11)$$

The finite volume formulation reduces to

$$\frac{d\bar{u}_i}{dt} = -\frac{1}{\Delta x} (F_{i+1/2} - F_{i-1/2}) + \bar{S}_i \quad (12)$$

where $F_{i\pm 1/2}$ are numerical fluxes at the cell interfaces. This equation shows that the solution update depends entirely on the imbalance of flux entering and leaving the control volume.

7 Fully Discrete Form

Using Forward Euler time integration gives

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x} (F_{i+1/2} - F_{i-1/2}) + \Delta t S_i \quad (13)$$

This is the standard first-order finite volume update equation.

8 Numerical Fluxes

At cell interfaces, the left and right cells generally possess different solution values. The physical flux at the interface therefore becomes ambiguous, especially in the presence of discontinuities. A numerical flux function is introduced to resolve this issue:

$$F_{i+1/2} = \hat{F}(u_L, u_R) \quad (14)$$

where u_L and u_R are the left and right states at the interface. Examples of numerical fluxes include:

$$\text{Rusanov, Roe, Godunov, HLL} \quad (15)$$

The numerical flux is one of the most important components of a finite volume solver.

9 Burgers' Equation

The inviscid Burgers' equation is

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \quad (16)$$

Its conservative form is

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left(\frac{u^2}{2} \right) = 0 \quad (17)$$

The physical flux is therefore

$$f(u) = \frac{u^2}{2} \quad (18)$$

The finite volume discretization becomes

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x} (F_{i+1/2} - F_{i-1/2}) \quad (19)$$

where the interface fluxes approximate the Burgers flux.

10 Rusanov Flux

One of the simplest numerical fluxes for Burgers' equation is the Rusanov flux:

$$F_{i+1/2} = \frac{1}{2} [f(u_L) + f(u_R)] - \frac{1}{2} a (u_R - u_L) \quad (20)$$

where

$$a = \max(|u_L|, |u_R|) \quad (21)$$

The first term represents a central average of the physical fluxes, while the second term introduces numerical dissipation for stability.

11 Method of Lines

The finite volume method naturally leads to a method-of-lines formulation:

$$\frac{du}{dt} = R(u) \quad (22)$$

where $R(u)$ represents the discrete spatial operator. Time integration methods such as Forward Euler, Runge-Kutta schemes, or SSPRK methods may then be applied independently of the spatial discretization.